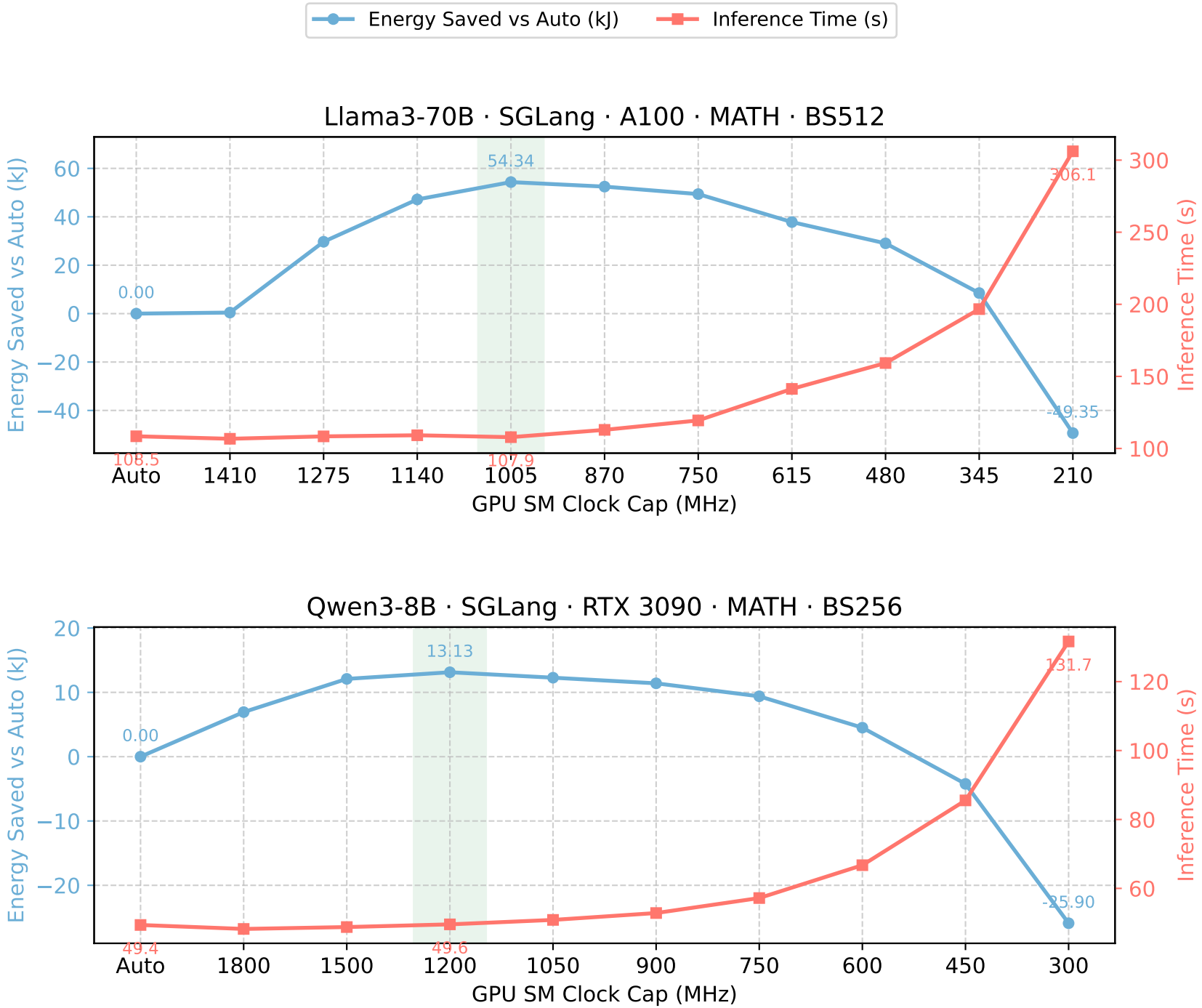

Shading: lowest energy under ≤5% time increase. A100 output2: Stage0–6 capped, Stage7 Auto, one run; Qwen3: two-run means.